

Differential Geometry

Exercise 9

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9.1 Problem

- (1) $f: M \rightarrow N$ be a local isometry, and denote the Riemannian curvature of M, N by Rm_M, Rm_N . For $p \in M$ and $X, Y, Z, W \in T_p M$ show that

$$\text{Rm}_M(X, Y, Z, W) = \text{Rm}_N(df(X), df(Y), df(Z), df(W)).$$

Equivalently, $\text{Rm}_M = f^* \text{Rm}_N$.

- (2) Let M be either the n -sphere of radius r or hyperbolic space H_{-1/r^2}^n . Show that

$$\text{Rm}_M(X, Y, Z, W) = \pm \frac{1}{r^2} (\langle X, Z \rangle \langle Y, W \rangle - \langle X, W \rangle \langle Y, Z \rangle),$$

and deduce that the sphere has positive sectional curvature $1/r^2$ and hyperbolic space has negative sectional curvature $-1/r^2$.

- (3) Show that H_{-1/r^2}^n is geodesically complete.

- (1) Since Riemannian curvature is local we can assume that f is a global isometry. Otherwise, we restrict our attention to open subsets $p \in U \subseteq M$ and $V \subseteq N$ such that f restricts to an isometry $U \rightarrow V$.

For an isometry, we know that the pullback of the Levi-Civita connection on N is the Levi-Civita connection on M . So if we let ∇^M, ∇^N be the connections on M, N then

$$\nabla_X^M Y = (f^{-1})_* \nabla_{f_* X}^N f_* Y.$$

So if we let R_M, R_N be the curvature tensors of M, N then for vector fields $X, Y, Z, W \in \mathfrak{X}(M)$,

$$\begin{aligned} R_M(X, Y)Z &= \nabla_Y^M \nabla_X^M Z - \nabla_X^M \nabla_Y^M Z + \nabla_{[X, Y]}^M Z \\ &= (f^{-1})_* (\nabla_{f_* Y}^N \nabla_{f_* X}^N f_* Z - \nabla_{f_* X}^N \nabla_{f_* Y}^N f_* Z + \nabla_{[f_* X, f_* Y]}^N f_* Z) \\ &= (f^{-1})_* R_N(f_* X, f_* Y) f_* Z. \end{aligned}$$

Thus

$$\begin{aligned} \text{Rm}_M(X, Y, Z, W) &= \langle R_M(X, Y)Z, W \rangle = \langle (f^{-1})_* R_N(f_* X, f_* Y) f_* Z, W \rangle \\ &= \langle R_N(f_* X, f_* Y) f_* Z, f_* W \rangle \\ &= \text{Rm}_N(f_* X, f_* Y, f_* Z, f_* W), \end{aligned}$$

and pointwise this means for tangent vectors $X, Y, Z, W \in T_p M$

$$\text{Rm}_M(X, Y, Z, W) = \text{Rm}_N(df(X), df(Y), df(Z), df(W)).$$

- (2) First we claim that it is sufficient to prove this identity at a single point $p \in M$. We will show that isometries act transitively on M , so for any $p, q \in M$ there is an isometry f of M such that $f(p) = q$. Since an isometry is in particular a diffeomorphism df_p is an isomorphism from $T_p M$ to $T_q M$, so for any $X, Y, Z, W \in T_q M$ we have by the first point

$$\text{Rm}_M(X, Y, Z, W) = \text{Rm}_M(df^{-1}X, df^{-1}Y, df^{-1}Z, df^{-1}W),$$

which by assumption is equal to

$$\begin{aligned} \pm \frac{1}{r^2} (\langle df^{-1}X, df^{-1}Z \rangle \langle df^{-1}Y, df^{-1}W \rangle - \langle df^{-1}X, df^{-1}W \rangle \langle df^{-1}Y, df^{-1}Z \rangle) \\ = \pm \frac{1}{r^2} (\langle X, Z \rangle \langle Y, W \rangle - \langle X, W \rangle \langle Y, Z \rangle). \end{aligned}$$

where the final equality is due to f^{-1} being an isometry.

So we need to show two things: that isometries act transitively and that the identity holds at a single point of our choosing. The isometries of M contain either $O(n+1)$ (for the sphere) or $O_+(1, n)$ (for hyperbolic space). $O(n+1)$ clearly acts transitively, as for any two points on the sphere we can normalize them and extend them to orthonormal bases of $\mathbb{R}^{n+1}$ and define an orthogonal transformation from one base to the other. This maps our first vector to the other.

For hyperbolic space, we do essentially the same thing. For $x, y \in H_{-1/r^2}^n$ normalize them to $\hat{x} = x/r$ and $\hat{y} = y/r$ so that $(\hat{x}, \hat{x}) = (\hat{y}, \hat{y}) = -1$. Now extend $\hat{x}, \hat{y}$ to orthonormal bases $\hat{x}, \bar{x}$ and $\hat{y}, \bar{y}$ and define a linear transformation $\hat{x} \mapsto \hat{y}$ and $\bar{x} \mapsto \bar{y}$. Since this is a linear map between orthonormal bases, it preserves the bilinear form and thus is in $O(1, n)$. Furthermore it preserves the sign of the first component of $\hat{x}$, since $\hat{x}$ and $\hat{y}$'s first component is both positive, and as we showed in the previous homework this implies the map is in $O_+(1, n)$. By linearity it maps x to y as desired. So we have shown isometries act transitively in both cases.

Now consider the map $f: \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$f(x) = \sqrt{r^2 \mp |x|^2},$$

where we potentially restrict the domain to the open subset ball of radius r around 0 so that this is smooth. The graph of f is the embedded submanifold of $\mathbb{R}^{n+1}$ given by

$$\text{graph}(f) = \left\{ (\sqrt{r^2 \mp |x|^2}, x) \mid x \in \mathbb{R}^n \right\}.$$

This is contained in M (when M is the sphere we take $-$ in the definition of f and $+$ for hyperbolic space) and is thus an embedded submanifold of M (as both are embedded in $\mathbb{R}^{n+1}$). Let f_i denote the i th partial derivative of f , so that

$$f_i = \mp \frac{x_i}{\sqrt{r^2 \mp |x|^2}}.$$

So we see that $x = 0$ is a critical point of f .

Let G denote either the Euclidean metric (when M is the sphere) or the Lorentzian metric (when M is hyperbolic) of $\mathbb{R}^{n+1}$. The metric on M is the pullback of G along the inclusion, and let us denote by g the pullback of this to $\text{graph}(f)$, which is just the pullback of G . We showed in the previous homework that in a small enough neighborhood of 0, relative to the coordinates $\mathbf{x}(x) = (f(x), x)$,

$$g_{ij} = \delta_{ij} \pm f_i f_j,$$

and when $x = 0$ we have

$$R_{ijk\ell} = \pm (f_{ik} f_{j\ell} - f_{i\ell} f_{jk}).$$

We are going to show that at $x = 0$, so the point $p = (r, 0)$ on M , the Riemannian curvature satisfies the given identity, and as explained this will be sufficient to conclude the identity for all of the points on M . It is further sufficient to show this for a basis of $T_p M$ since Rm is tensorial and thus multilinear, and so is the right-hand side of the identity. So we demonstrate the identity via the local coordinate frame given by $\mathbf{x}$, that is we want to show when $x = 0$,

$$R_{ijk\ell} = \pm \frac{1}{r^2} (g_{ik} g_{j\ell} - g_{i\ell} g_{jk}).$$

Recalling that at $x = 0$ we have $g_{ij} = \delta_{ij}$, to do this it is sufficient to show that

$$f_{ij} = \mp \frac{1}{r} \delta_{ij}.$$

Indeed, for $i \neq j$ we have

$$f_{ij} = \mp \frac{x_i x_j}{(r^2 - |x|^2)^{3/2}},$$

which is zero when $x = 0$. And

$$f_{ii} = \mp \frac{r^2 - |x|^2 + x_i^2}{(r^2 - |x|^2)^{3/2}},$$

which at $x = 0$ is equal to $\mp 1/r$ as desired.

- (3) We claim that for any Riemannian manifold M on which its isometries act transitively, M is geodesically complete. Let $p \in M$, then we know $\exp_p$ is defined at least on some ball $B_\varepsilon(0) \subseteq T_p M$ of radius ε (taken relative to the Riemannian metric). Meaning for $v \in B_\varepsilon(0)$, γ_v is defined on an open interval containing 0 and 1. We claim that γ_v is actually defined on all of $\mathbb{R}$ for all $v \in B_\varepsilon(0)$. Assume not, so there is a geodesic whose domain is bound, we assume without loss of generality it is bound from above. So let

$$a = \inf \{t \in \mathbb{R}_{>0} \mid \text{There is a } v \in B_\varepsilon(0) \text{ such that } \gamma_v \text{ is not defined at } t\},$$

which is finite by assumption and at least 1, so a is positive. So for all $v \in B_\varepsilon(0)$ we have that γ_v is defined on at least $[0, a)$.

For $v \in B_\varepsilon(0)$, let $q = \gamma_v(a/2)$ and $w = \dot{\gamma}_v(a/2)$. Since γ_v is a geodesic,

$$|w|_g = |\dot{\gamma}_v(a/2)|_g = |\dot{\gamma}_v(0)| = |v|_g < \varepsilon.$$

Let f be an isometry of M such that $f(q) = p$, and let $u = df(w)$. Since f is an isometry, $|u| = |w| = |v| < \varepsilon$. Thus γ_u is defined on at least $[0, a)$.

Now consider $f^{-1} \circ \gamma_u$, which is a geodesic since f is an isometry. And

$$f^{-1} \circ \gamma_u(0) = f^{-1}(p) = q, \quad (f^{-1} \circ \gamma_u)'(0) = df_q^{-1}(u) = w.$$

By uniqueness, $f^{-1} \circ \gamma_u$ and γ_w agree on their common domains. Since $w = \dot{\gamma}_v(a/2)$ we have that again by uniqueness, $\gamma_w(t) = \gamma_v(t + a/2)$. Thus

$$f^{-1} \circ \gamma_u(t) = \gamma_v(t + a/2)$$

when both sides are defined, which contains $[0, a/2)$. But then consider

$$\alpha(t) = \begin{cases} \gamma_v(t) & t < a \\ f^{-1} \circ \gamma_u(t - a/2) & a/2 < t < 3a/2 \end{cases}.$$

Since both definitions agree on the intersection of their domains, this is a smooth curve, and as both curves are geodesics it is a geodesic. So α is the geodesic with initial velocity v , and since it is defined up until $3a/2$ we see that all geodesics with initial velocity in $B_\varepsilon(0)$ are defined up until $3a/2$, contradicting the definition of a .

Thus all geodesics in $B_\varepsilon(0)$ are defined on all of $\mathbb{R}$. Now let $v \in T_p M$, then we know that for $c \in \mathbb{R}$

$$\gamma_{cv}(t) = \gamma_v(ct)$$

whenever either side is defined. And in particular if γ_{cv} is defined on all of $\mathbb{R}$, so is γ_v . There is a c such that $cv \in B_\varepsilon(0)$ (e.g. $c = |v|_g + 1$), and we just showed that γ_{cv} is defined on all of $\mathbb{R}$, so that means γ_v is too. Thus M is geodesically complete.

9.2 Problem

Let J be a Jacobi field along γ . Show that there is a family of geodesics γ_s such that $\gamma = \gamma_0$ and $J = \left. \frac{\partial \gamma_s}{\partial s} \right|_{s=0}$.

Any family of geodesics will be of the form

$$\gamma_s(t) = \exp_{\alpha(s)}(tX(s))$$

where α is a smooth curve and X is a vector field along α (so $X(s) \in T_{\alpha(s)}M$ and this is defined). So let us define

$$\Gamma(s, t) = \exp_{\alpha(s)}(tX(s)),$$

and we will search for α, X that satisfy our conditions. First of all we have

$$\Gamma(0, t) = \exp_{\alpha(0)}(tX(0)),$$

so we know that we must have $\alpha(0) = \gamma(0)$ and $X(0) = \dot{\gamma}(0)$.

Since Γ is a smooth family of geodesics by construction, its variation field $V(t) = \partial_s \Gamma(0, t)$ is a Jacobi field along γ . We want to find suitable α, X such that $V = J$. By uniqueness of Jacobi fields it is sufficient to show $J(0) = V(0)$ and $D_t J(0) = D_t V(0)$.

Computing shows

$$\partial_s \Gamma(0, 0) = \left. \frac{\partial}{\partial s} \right|_{s=0} \Gamma(s, 0) = \left. \frac{\partial}{\partial s} \right|_{s=0} \exp_{\alpha(s)}(0) = \left. \frac{\partial}{\partial s} \right|_{s=0} \alpha(s) = \dot{\alpha}(0).$$

So we must have $\dot{\alpha}(0) = J(0)$. Now by symmetry

$$D_t V(0) = D_t \partial_s \Gamma(0, 0) = D_s \partial_t \Gamma(0, 0).$$

And

$$\partial_t \Gamma(s, 0) = \partial_t \exp_{\alpha(s)}(tX(s)) = X(s),$$

so we want

$$D_t J(0) = D_s X(0).$$

So our conditions are

$$\alpha(0) = \gamma(0), \quad X(0) = \dot{\gamma}(0), \quad \dot{\alpha}(0) = J(0), \quad D_s X(0) = D_t J(0).$$

If we take local coordinates around $p = \gamma(0)$ these become

$$\alpha(0) = 0, X^i(0) = \dot{\gamma}^i(0), \quad \dot{\alpha}^i(0) = J^i(0), \quad \dot{X}^i(0) + X^j(0)\dot{\alpha}^k(0)\Gamma_{kj}^i(p) = \dot{J}^i(0) + J^j(0)\dot{\gamma}^k(0)\Gamma_{kj}^i(p).$$

The final condition can be rewritten, by the previous conditions, as

$$\dot{X}^i(0) + \dot{\gamma}^j(0)J^k(0)\Gamma_{kj}^i(p) = \dot{J}^i(0) + J^j(0)\dot{\gamma}^k(0)\Gamma_{kj}^i(p),$$

and by symmetry, $\Gamma_{kj}^i = \Gamma_{jk}^i$ so that this is equivalent to

$$\dot{X}^i(0) = \dot{J}^i(0).$$

So we can define, in these coordinates,

$$\alpha(t) = (J^1(0)x_1, \dots, J^n(0)x_n), \quad X(t) = (J^i(t) + \dot{\gamma}^i(0) - J^i(0))\partial_i|_{\alpha}.$$

And these clearly satisfy the conditions.

9.3 Problem (DoCarmo, Riemannian Geometry, page 77 problem 1)

Let (u, v) denote the Cartesian coordinates of $\mathbb{R}^2$. Further let $f, g: (v_0, v_1) \rightarrow \mathbb{R}$ be smooth functions defined on an open interval such that $f'(v)^2 + g'(v)^2 \neq 0$ and $f(v) \neq 0$. Define

$$U = \{(u, v) \in \mathbb{R}^2 \mid u_0 < u < u_1, v_0 < v < v_1\}, \quad \varphi: U \rightarrow \mathbb{R}^3, \quad (u, v) \mapsto (f(v) \cos u, f(v) \sin u, g(v)).$$

- (1) Show that φ is an immersion, and its image $\varphi(U)$ is called a **surface of revolution** S . The image under φ of the constant curves $u = c, v = c$ are called the **meridians** and **parallels** of S respectively.
- (2) Show that the induced metric in the coordinates (u, v) is given by

$$g_{11} = f^2, \quad g_{12} = 0, \quad g_{22} = (f')^2 + (g')^2.$$

- (3) Show that the geodesic equation becomes the system

$$\begin{aligned} \frac{d^2 u}{dt^2} + \frac{2ff'}{f^2} \frac{du}{dt} \frac{dv}{dt} &= 0 \\ \frac{d^2 v}{dt^2} - \frac{ff'}{(f')^2 + (g')^2} \left(\frac{du}{dt} \right)^2 + \frac{f'f'' + g'g''}{(f')^2 + (g')^2} \left(\frac{dv}{dt} \right)^2 &= 0. \end{aligned}$$

- (4) Deduce the following geometric meaning of the geodesic equations above. The second equation means that, except for meridians and parallels, that $|\gamma'(t)|^2$ is constant along $\gamma(t)$. The first equation means that if $\beta(t) < \pi$ is the angle obtained at the intersection of γ with a parallel P intersecting γ at $\gamma(t)$, then $r \cos \beta$ is constant, where r is the radius of the parallel P .
The equation $r \cos \beta = \text{const}$ is called **Clairut's relation**.
- (5) Use Clairut's relation to show that a geodesic of the paraboloid, so $f(v) = v, g(v) = v^2$ with $0 < v < \infty$ and $-\varepsilon < u < 2\pi + \varepsilon$, which is not a meridian must intersect itself an infinite number of times.

- (1) We compute the Jacobian of φ :

$$D\varphi = \begin{pmatrix} f'(v) \cos u & -f(v) \sin u \\ f'(v) \sin u & f(v) \cos u \\ g'(v) & 0 \end{pmatrix}.$$

Since $f(v)$ is nonzero, the right column is nonzero, so if $g'(v) \neq 0$ then the two columns are linearly independent. Otherwise $g'(v) = 0$ so $f'(v) \neq 0$ and the determinant of the upper square matrix is

$$f'(v)f(v) \cos^2(u) + f'(v)f(v) \sin^2(u) = f'(v)f(v),$$

which is nonzero. So in all cases, $D\varphi$ has rank 2 and is thus an immersion.

- (2) If we let ∂_i denote the global coordinate frame of $\mathbb{R}^n$, then the coordinate frame of S is given by $\tilde{\partial}_i = d\varphi(\partial_i)$ which in terms of the ∂_j of $\mathbb{R}^3$ is

$$\tilde{\partial}_i = \frac{\partial \varphi^j}{\partial \partial_i} \partial_j,$$

so these are

$$\begin{aligned}\tilde{\partial}_v &= f'(v) \cos(u) \partial_1 + f'(v) \sin(u) \partial_2 + g'(v) \partial_3 \\ \tilde{\partial}_u &= -f(v) \sin(u) \partial_1 + f(v) \cos(u) \partial_2.\end{aligned}$$

Then $\tilde{g}_{ij} = g(\tilde{\partial}_i, \tilde{\partial}_j)$ is given by

$$\begin{aligned}\tilde{g}_{vv} &= f'(v)^2 \cos^2(u) + f'(v)^2 \sin^2(u) + g'(v)^2 = f'(v)^2 + g'(v)^2 \\ \tilde{g}_{uv} &= -f'(v)f(v) \cos(u) \sin(u) + f'(v)f(v) \cos(u) \sin(u) = 0 \\ \tilde{g}_{uu} &= f(v)^2 \sin^2(u) + f(v)^2 \cos^2(u) = f(v)^2,\end{aligned}$$

as desired.

(3) Let us order the coordinate chart by (u, v) . The Christoffel symbols are given by

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\tilde{\partial}_i g_{j\ell} + \tilde{\partial}_j g_{i\ell} - \tilde{\partial}_\ell g_{ij}).$$

Since g_{ij} is the diagonal matrix

$$g = \begin{pmatrix} f(v)^2 & 0 \\ 0 & f'(v)^2 + g'(v)^2 \end{pmatrix} \Rightarrow g^{-1} = \begin{pmatrix} f(v)^{-2} & 0 \\ 0 & \frac{1}{f'(v)^2 + g'(v)^2} \end{pmatrix},$$

we see that

$$\Gamma_{ij}^k = \frac{1}{2} g^{kk} (\tilde{\partial}_i g_{jk} + \tilde{\partial}_j g_{ik} - \tilde{\partial}_k g_{ij}).$$

Now g_{ij} is either zero, for $i \neq j$, or a function of v . Thus $\partial_1 g_{ij}$ will always be zero, and $\partial_k g_{ij} = 0$ if $i \neq j$. Now computing gives

$$\begin{aligned}\Gamma_{11}^1 &= \Gamma_{22}^1 = \Gamma_{11}^2 = 0 \\ \Gamma_{12}^1 &= \frac{f'}{f} \\ \Gamma_{11}^2 &= -\frac{f'f}{(f')^2 + (g')^2} \\ \Gamma_{22}^2 &= \frac{f'f'' + g'g''}{(f')^2 + (g')^2}.\end{aligned}$$

Plugging these into the geodesic equation $\ddot{x}^k + \dot{x}^i \dot{x}^j \Gamma_{ij}^k = 0$ we get the two equations

$$\begin{aligned}\frac{d^2 u}{dt^2} + 2 \frac{du}{dt} \cdot \frac{dv}{dt} \cdot \frac{f'}{f} &= 0 \\ \frac{d^2 v}{dt^2} - \frac{f'f}{(f')^2 + (g')^2} \left(\frac{du}{dt} \right)^2 + \frac{f'f'' + g'g''}{(f')^2 + (g')^2} \left(\frac{dv}{dt} \right)^2 &= 0,\end{aligned}$$

as desired.

(4) The energy of a geodesic is given by

$$|\gamma'(t)|^2 = \left| \dot{u} \tilde{\partial}_1 + \dot{v} \tilde{\partial}_2 \right|^2 = f^2 \dot{u}^2 + ((f')^2 + (g')^2) \dot{v}^2.$$

Differentiating this gives (the derivative of $f = f(v(t))$ is $\dot{v}f'$),

$$2(f\dot{v}f'\dot{u}^2 + f^2\dot{u}\ddot{u} + (f'\dot{v}f'' + g'\dot{v}g'')\dot{v}^2 + ((f')^2 + (g')^2)\dot{v}\ddot{v}).$$

We want to show that this is zero. The second geodesic equation gives us, after multiplying by $\dot{v}((f')^2 + (g')^2)$ that the above equation is equal to (after dividing by 2)

$$f f' \dot{v} \dot{u}^2 + f^2 \dot{u} \ddot{u} + f f' \dot{u}^2 \dot{v}.$$

We can divide by f , so we want to show

$$f' \dot{v} \dot{u}^2 + f \dot{u} \ddot{u} + f' \dot{u}^2 \dot{v} = 0,$$

It is sufficient to show that

$$f \ddot{u} + 2f' \dot{v} \dot{u} = 0.$$

This is due to the first equation, and so the derivative of $|\gamma'(t)|^2$ is zero and thus it is constant.

If $P = (\hat{u}, c)$ is a parallel which intersects γ at $\gamma(t)$ then the cosine of the angle between them, $\beta(t)$, is given by

$$\cos \beta(t) = \frac{\langle P', \gamma' \rangle}{|P'| |\gamma'|},$$

and since $P' = \hat{u}' \tilde{\partial}_1$ and $\gamma' = \dot{u} \tilde{\partial}_1 + \dot{v} \tilde{\partial}_2$ we see that this is equal to

$$\cos \beta(t) = \frac{\hat{u}' \dot{u} f^2}{\hat{u}' f |\gamma'|} = \frac{f \dot{u}}{|\gamma'|}.$$

The radius of P is $f(c) = f(v)$, so Clairut's equation says that

$$r \cos \beta(t) = \frac{f^2 \dot{u}}{|\gamma'|}$$

is constant. Since $|\gamma'|$ is constant as γ' is a geodesic, we need to show $f^2 \dot{u}$ is constant. Differentiating gives (recall that the derivative of $f = f(v)$ is $\dot{v} f'$)

$$(f^2 \dot{u})' = 2f' f \dot{v} \dot{u} + f^2 \ddot{u},$$

the first equation means $f^2 \ddot{u} = -2f' f \dot{v} \dot{u}$ so this is zero as desired.

- (5) Let γ be a non-meridian geodesic on the paraboloid, so $r \cos \beta$ is constant by Clairut, and by above this means $f \dot{u}$ is constant. Since u is not constant, that $\dot{u}$ is not identically zero and thus $f \dot{u}$ is not identically zero, and since it is a constant this means that $f \dot{u} \neq 0$ always so $\dot{u} \neq 0$ always. Thus $\dot{u}$ is either always positive or always negative, so by potentially reparameterizing we can assume $\dot{u} > 0$.

Now in our case, $f(v) = v$ and $g(v) = v^2$ so

$$g_{11} = v^2, \quad g_{12} = 0, \quad g_{22} = 1 + 4v^2.$$

And Clairut's relation is

$$v \cos \beta(t) = c,$$

since the radius is $f(v) = v$. Let us reparameterize γ so that it has unit speed: $|\gamma'| = 1$; this does not affect the claim. We know that

$$\cos \beta(t) = v \dot{u},$$

so we get that

$$\dot{u} = \frac{c}{v^2}.$$

Now,

$$1 = |\gamma'|^2 = \dot{u}^2 v^2 + \dot{v}^2 (1 + 4v^2) \geq 4\dot{v}^2 v^2,$$

and so $2\dot{v}v = (v^2)' \leq 1$. Integrating shows that $v^2 \leq t + v(0)^2$, and so

$$\dot{u} \geq \frac{c}{t + v(0)^2},$$

and integrating shows that u approaches ∞ as $t \rightarrow \infty$, since the right-hand side's integral diverges to infinity. Thus u strictly increases to infinity as $t \rightarrow \infty$.

Now we want to show that $v \rightarrow \infty$ as $t \rightarrow \pm\infty$. We have that $v = c/\cos\beta$ and so $v \geq |c|$, and this is an equality iff $\cos\beta = \pm 1$ iff $v\dot{u} = \pm 1$ iff $v^2\dot{u}^2 = 1$. Since $1 = \dot{u}^2v^2 + \dot{v}^2(1 + 4v^2)$, this occurs iff $\dot{v} = 0$.

Now suppose $\dot{v}(t_1) = \dot{v}(t_2) = 0$ with $t_1 < t_2$, then $v(t_1) = v(t_2) = |c|$, and if we consider the interval $[t_1, t_2]$ then v must have zero derivative somewhere in it by Rolle's theorem. But this means that there is a $t_0 \in (t_1, t_2)$ such that $v(t_0) = |c|$ since $\dot{v}(t_0) = 0$. So between every two points where $v(t_1) = v(t_2) = |c|$, there is a third which satisfies this. Since v is smooth, this means that v is identically equal to $|c|$ on $[t_1, t_2]$. But $v = |c|$ implies that $\dot{u} = c/c^2 = 1/c$ is constant and thus $u = t/c$. But this clearly doesn't satisfy the second geodesic equation, since $\dot{u}^2$ is nonzero but all the other terms are.

So $\dot{v}$ is only zero at most at a single point. Suppose this point is t_0 , so that $\dot{v}$ has the same sign on (t_0, ∞) (if t_0 doesn't exist, take $t_0 = -\infty$). Now if $v(t)$ doesn't approach infinity, then since it is monotonic for $t > t_0$ this means it must converge to some value $\lim v(t) = L < \infty$ and $v(t)$ is bound. We know Since $v(t)$ converges to a finite value at infinity, $\lim_t \dot{v}$ must be zero. The second geodesic equation gives

$$\ddot{v} - \frac{v\dot{u}^2}{1 + 4v^2} + \frac{4v\dot{v}^2}{1 + 4v^2} = 0,$$

which means

$$\ddot{v} = \frac{c^2}{v^3(1 + 4v^2)} - \frac{4v\dot{v}^2}{1 + 4v^2},$$

and as $t \rightarrow \infty$ this converges to $c^2/(L^3(1 + 4L^2)) > 0$ so $\ddot{v}$ is eventually greater than a positive constant. Thus $\dot{v}$ must increase to infinity, which is a contradiction to it having limit zero at infinity.

So $\lim v(t) = \infty$ and since $\dot{v}$ has the same sign, for $t > t_0$ this means that it must have positive sign, and thus v increases to infinity as $t \rightarrow \infty$. Thus $v(t) \rightarrow \infty$ as $t \rightarrow \infty$, and by reparameterizing $\gamma(-t)$ we see that $v(t) \rightarrow \infty$ as $t \rightarrow -\infty$, and this convergence is also monotone. Note that this shows that this shows there must be some t_0 where $\dot{v}(t_0) = 0$, as otherwise we would have to have $v(t)$ always increasing and then we couldn't have that $v(t) \rightarrow \infty$ as $t \rightarrow -\infty$.

Thus v is increasing to infinity on (t_0, ∞) and decreasing from infinity on $(-\infty, t_0)$. In particular v is a diffeomorphism from each interval to $(|c|, \infty)$, and since u also goes to infinity we can write u in terms of v on each interval. That is, $u = u_1(v)$ for $t > t_0$ and $u = u_2(v)$ for $t < t_0$ (or for large/small enough t). Since u is increasing, and v is decreasing on $(-\infty, t_0)$ we see that $u_2(v)$ is a decreasing function of v , and $u_1(v)$ is an increasing function. Thus $u_1(v) - u_2(v)$ is increasing to infinity, and thus there must be infinitely many v such that $u_1(v) - u_2(v)$ is a positive multiple of 2π . For each of those values of v we have that $\cos u_1 = \cos u_2$ and $\sin u_1 = \sin u_2$, so γ intersects itself at (u_1, v) and (u_2, v) , and thus infinitely many times.

9.4 Problem

(1) Let (M, g) be Riemannian, and let $f: M \rightarrow M$ be an isometry. Show that

$$\text{Fix}(f) = \{p \in M \mid f(p) = p\}$$

is a submanifold of M .

- (2) Let $f: M \rightarrow M$ be a diffeomorphism of finite order, so $f^k = \text{id}$ for some k . Show that $\text{Fix}(f)$ is a submanifold.
- (3) Show that the previous point is not true if f doesn't have finite order.

- (1) For $p \in \text{Fix}(f)$ there is an $r > 0$ such that $\exp_p$ is a diffeomorphism from $B_r(0) \subseteq T_p M$. Let $W \subseteq T_p M$ be the subspace fixed by df_p . Then we claim that $\exp_p$ forms a diffeomorphism $W \cap B_r(0) \rightarrow \text{Fix}(f) \cap \exp_p(B_r(0))$. Since $\exp_p$ is a diffeomorphism from $B_r(0)$ we need only show that it maps into $\text{Fix}(f) \cap \exp_p(B_r(0))$ and does so surjectively.

For $v \in W \cap B_r(0)$ we need to show that $\exp_p(v)$ is fixed by f . Since $\exp_p(tv)$ is the unique geodesic starting at p with initial velocity v , $f \circ \exp_p(tv)$ is a geodesic too as f is an isometry. It starts at $f(\exp_p(0)) = f(p) = p$ with initial velocity

$$(f \circ \exp_p(tv))'(0) = df_p(v) = v,$$

so by uniqueness $f \circ \exp_p(tv) = \exp_p(tv)$ and in particular for $t = 1$ we see that $\exp_p(v)$ is fixed by f as desired, meaning $\exp_p$ does indeed map into $\text{Fix}(f) \cap \exp_p(B_r(0))$.

Conversely for $q = \exp_p(v) \in \text{Fix}(f) \cap \exp_p(B_r(0))$ we claim that $v \in B_r(0)$ is fixed by df . As before, $f \circ \exp_p(tv)$ is a geodesic with initial point p and initial velocity $df_p(v)$. Since f is an isometry it preserves the norm, so $df_p(v) \in B_r(0)$. This means that $f \circ \exp_p(tv) = \exp_p(df_p(v))$ by uniqueness of geodesics, and it is defined at $t = 1$ which gives

$$\exp_p(df_p(v)) = f(\exp_p(v)) = \exp_p(v),$$

and since $\exp_p$ is injective on $B_r(0)$ we get $df_p(v) = v$ so $v \in W$. Thus $\exp_p$ is a bijection between Euclidean space $W \cap B_r(0)$ and $\text{Fix}(f) \cap \exp_p(B_r(0))$, we use these as charts for $\text{Fix}(p)$ (since $\exp_p(B_r(0))$ is open in M , $\text{Fix}(f) \cap \exp_p(B_r(0))$ is open in the subspace topology).

These charts are smoothly compatible, since $\exp_p$ and is a diffeomorphism onto its image in M . Thus this makes $\text{Fix}(f)$ a submanifold of M .

- (2) Let g_0 be an arbitrary metric on M , define a new metric g by

$$g(X, Y) = \frac{1}{k} \sum_{i=0}^{k-1} g_0(df^i(X), df^i(Y)).$$

Since g_0, df are smooth, so too is g , so this defines a tensor field. It is also positive definite at each point since $g_0(X, X)$ is nonnegative and $g(X, X) = 0$ iff $g_0(df^i(X), df^i(X)) = 0$ for all i , iff $df^i(X) = 0$ iff $X = 0$. And f is an isometry since

$$g(df(X), df(Y)) = \frac{1}{k} \sum_{i=0}^{k-1} g_0(df^{i+1}(X), df^{i+1}(Y)),$$

and by reordering the indices since $df^k = d(f^k) = d(\text{id}) = \text{id}$, we see this is equal to $g(X, Y)$.

So f is an isometry of (M, g) and by the previous point $\text{Fix}(f)$ is a submanifold of M .

- (3) Consider the function

$$h: \mathbb{R} \rightarrow \mathbb{R}, \quad h(x) = \exp\left(-\frac{1}{x^2}\right) \sin\left(\frac{1}{x}\right).$$

This is clearly smooth at points other than 0, and computing its first derivative gives

$$h'(x) = \exp\left(-\frac{1}{x^2}\right) \left(\frac{2}{x^3} \sin\left(\frac{1}{x}\right) - \frac{1}{x^2} \cos\left(\frac{1}{x}\right) \right),$$

and inductively we see that

$$h^{(n)}(x) = \exp\left(-\frac{1}{x^2}\right)(P_n(1/x) \sin(1/x) + Q_n(1/x) \cos(1/x)),$$

for polynomials P_n, Q_n . This converges to 0 at $x = 0$, so h is smooth over all of $\mathbb{R}$. The inductive step is just a computation: $h^{(n+1)}(x)$ is equal to

$$\begin{aligned} & 2x^{-3} \exp(-x^{-2}) \cdot (P_n(1/x) \sin(1/x) + Q_n(1/x) \cos(1/x)) \\ & - x^{-1} \exp(-x^{-2}) (P'_n(1/x) \sin(1/x) + P_n(1/x) \cos(1/x) + Q'_n(1/x) \cos(1/x) - Q_n(1/x) \sin(1/x)) \\ & = \exp(-x^{-2}) \cdot (P_{n+1}(1/x) \sin(1/x) + Q_{n+1}(1/x) \cos(1/x)) \end{aligned}$$

for

$$P_{n+1}(x) = 2x^3 P_n(x) - xP'_n(x) + xQ_n(x), \quad Q_{n+1}(x) = 2x^3 Q_n(x) - xP_n(x) + xQ'_n(x).$$

Now consider the vector field given by $X = h(x)dx$, which is the same as ∇H where H is an antiderivative of the smooth function h .

We claim that X is a complete vector field. Notice that $|h(x)|$ is bound, e.g. by 1. So if γ is any maximal integral curve of X , so $\gamma'(t) = h(\gamma(t))$ then

$$\gamma(t) = \gamma(0) + \int_0^t \gamma'(s) ds = \gamma(0) + \int_0^t h(\gamma(s)) ds,$$

which implies that

$$|\gamma(t)| \leq |\gamma(0)| + \int_0^t |h(\gamma(s))| ds \leq |\gamma(0)| + |t|.$$

The escape lemma for integral curves says that if γ is maximal whose domain is bounded by b , then $\gamma[t_0, b)$ is not contained in a compact subset of M , so in $\mathbb{R}$'s case it is unbound. But as we showed, $\gamma[t_0, b)$ is bound, so γ 's domain cannot be bounded, and thus it must be all of $\mathbb{R}$. So X is complete, and so its flow θ is defined on all of $\mathbb{R} \times M \rightarrow M$.

Finally, let Φ be X 's time-one flow, i.e. $\Phi(x) = \theta(1, x)$. As flow is smooth, so is Φ . Furthermore, by the fundamental theorem of flows, Φ is a diffeomorphism $M \rightarrow M$ (since X is complete). But we claim that $\text{Fix}(\Phi) = Z(h)$, the set of h 's zeroes.

Indeed, if $h(x) \neq 0$ then $\theta^{(x)}(t) = \theta(t, x)$ is an immersion and thus has nonzero derivative (integral curves starting at regular points of the field are immersions). Since

$$\dot{\theta}^{(x)}(t) = h(\theta^{(x)}(t)),$$

this means that $h = H'$ is nonzero on the integral curve $\theta^{(x)}(t)$, and thus H is injective on it. If $\Phi(x) = x$ then this means $\theta^{(x)}(1) = \theta^{(x)}(0)$, and thus $H(\theta^{(x)}(1)) = H(\theta^{(x)}(0))$ contradicting h 's injectivity on the integral curve. For $h(x) = 0$, the integral curve starting at x is constant, $\theta^{(x)}(t) = x$, and in particular $\Phi(x) = x$. So

$$\text{Fix}(\Phi) = Z(h).$$

But

$$Z(h) = \left\{ \frac{1}{\pi k} \mid k \in \mathbb{Z} - 0 \right\} \cup \{0\},$$

which is a countable non-discrete subspace of $\mathbb{R}$ (since any neighborhood of 0 contains some other zero of h), and thus cannot be a submanifold (since countable manifolds are discrete).

9.5 Problem

Let M be a smooth manifold, and denote its space of differential k -forms by $\Omega^k(M)$, and let $n = \dim M$.

- (1) Let g be a Riemannian metric on M . Define an operator $\operatorname{div}_g: \mathfrak{X}(M) \rightarrow C^\infty(M)$ pointwise by

$$(\operatorname{div}_g X)(p) = \operatorname{tr}(\nabla X|_p),$$

where $\nabla X|_p: T_p M \rightarrow T_p M$ is the linear operator $v \mapsto \nabla_v X|_p$. Show that in a geodesic frame (E_i) at p and $X = f^i E_i$, one has

$$(\operatorname{div}_g X)(p) = (E_i f^i)(p).$$

- (2) Suppose M is orientable. Show that there exists a unique $\nu_g \in \Omega^n(M)$ called the **volume form** induced by g satisfying the equivalent conditions:

- (i) for every oriented local orthonormal frame (E_i) , $\nu_g(E_1, \dots, E_n) = 1$,
- (ii) in every oriented local coordinate coframe,

$$\nu_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n.$$

- (3) Given the volume form $\nu_g \in \Omega^n(M)$, let $\mathcal{L}_X \nu_g$ be its Lie derivative relative to a vector field X . Show that

$$\mathcal{L}_X \nu_g = (\operatorname{div}_g X) \nu_g.$$

- (1) Since $(E_i|_p)$ forms a basis of $T_p M$, we can compute the trace of $\nabla X|_p$ relative to $E_i|_p$. We have that

$$\nabla_{E_i|_p} X|_p = \nabla_{E_i} X|_p = \nabla_{E_i} f^j E_j|_p = (E_i f^j) E_j|_p + f^j \nabla_{E_i} E_j|_p,$$

and since E_i is a geodesic frame at p , this is equal to $(E_i f^j)(p) E_j|_p$. So relative to the basis $(E_i|_p)$, ∇X is represented by the matrix $A_j^i = E_i f^j(p)$, and thus its trace is

$$\operatorname{div}_g X(p) = \sum_i A_i^i = \sum_i E_i f^i(p),$$

as desired.

- (2) First we show the two conditions are equivalent. Let (E_i) be an oriented local orthonormal frame and (x^i) an oriented chart which induces the oriented local coordinate frame (∂_i) . Let A and B be the transition functions between them, so

$$E_i = A_i^j \partial_j, \quad \partial_j = B_j^i E_i$$

meaning A and B are inverse matrices. And we know for any n -form $\omega \in \Omega^n(M)$,

$$\omega(E_1, \dots, E_n) = \det(A_i^j) \omega(\partial_1, \dots, \partial_n), \quad \omega(\partial_1, \dots, \partial_n) = \det(B_j^i) \omega(E_1, \dots, E_n).$$

We will show that $\det(B_j^i) = \sqrt{\det(g_{ij})}$, which shows that if $\nu_g(\bar{E}) = 1$ then $\nu_g(\bar{\partial}) = \sqrt{\det(g_{ij})}$ and vice versa. Since both frames are oriented, their transition maps A, B have positive determinant, so it is sufficient to show $\det(B_j^i)^2 = \det(g_{ij})$. We know that

$$g_{ij} = \langle \partial_i, \partial_j \rangle = \langle B_i^k E_k, B_j^\ell E_\ell \rangle = B_i^k B_j^\ell \delta_{k\ell} = \sum_k B_i^k B_j^k.$$

So we have that

$$g_{ij} = (B^\top \cdot B)_j^i,$$

meaning $g = B^\top B$, so

$$\det(g) = \det(B^\top B) = \det(B^\top) \det(B) = \det(B)^2,$$

as desired. Now, $\nu_g(\partial_1, \dots, \partial_n) = \sqrt{\det(g)}$ is equivalent to $\nu_g = \sqrt{\det(g)} dx^1 \wedge \dots \wedge dx^n$ because the coefficient of $dx^1 \wedge \dots \wedge dx^n$ in the representation of ν_g is given by $\nu_g(\partial_1, \dots, \partial_n)$. So we have shown the equivalence of the two conditions.

Either of the two points uniquely define ν_g , since around any point we have a unique representation of ν_g in terms of a local coordinate coframe, or local orthonormal coframe.

Now we just need to show that ν_g is well-defined. We take the first point as the definition, meaning in terms of any oriented local orthonormal frame (E_i) with dual frame (ε^i) we define

$$\nu_g = \varepsilon^1 \wedge \dots \wedge \varepsilon^n.$$

To show that this is well-defined, we take another oriented local orthonormal frame (F_i) with dual frame (δ^i) and we want to show that

$$\varepsilon^1 \wedge \dots \wedge \varepsilon^n = \delta^1 \wedge \dots \wedge \delta^n.$$

Let A_j^i be the transition matrix, so $F_j = A_j^i E_i$ and thus $\varepsilon^j = A_i^j \delta^i$ so

$$\varepsilon^1 \wedge \dots \wedge \varepsilon^n = \det(A) \delta^1 \wedge \dots \wedge \delta^n.$$

Since A is the transition map between orthonormal bases, it is unitary and thus $|\det(A)| = 1$, since further both E_i, F_i are positively oriented, A has positive determinant and so $\det(A) = 1$ giving us the desired result. So ν_g is well-defined, as desired.

- (3) Let (E_i) be an oriented geodesic frame at p , and write $X = f^i E_i$ locally. Since the Lie derivative of a differential form is itself a differential form, it is sufficient to show that at p we have

$$\mathcal{L}_X(\nu_g)(E_1, \dots, E_n) = (\operatorname{div}_g X) \nu_g(E_1, \dots, E_n),$$

then the result also follows for any permutation of $E_1, \dots, E_n$ (by antisymmetry of differential forms) so that we get the result for all vectors $X_1, \dots, X_n \in T_p M$. This is true for all points p so we get the result globally.

For the remainder of this solution, everything is being evaluated at the point p implicitly.

The right-hand side is by definition just $\operatorname{div}_g X$. The left-hand side is, by the formula for the Lie derivative of covariant tensors,

$$\mathcal{L}_X(\nu_g)(E_1, \dots, E_n) = X(\nu_g(E_1, \dots, E_n)) - \sum_i \nu_g(E_1, \dots, [X, E_i], \dots, E_n).$$

Since $\nu_g(E_1, \dots, E_n) = 1$ is constant, the first term is zero. By symmetry of the Levi-Civita connection,

$$\nabla_X E_i - \nabla_{E_i} X = [X, E_i],$$

The first term on the left is $f^j \nabla_{E_j} E_i = 0$, so we see

$$[X, E_i] = -\nabla_{E_i} X = -\nabla_{E_i} f^j E_j = -(E_i f^j) E_j.$$

Thus

$$\mathcal{L}_X(\nu_g)(E_1, \dots, E_n) = \sum_i \nu_g(E_1, \dots, (E_i f^j) E_j, \dots, E_n) = \sum_i (E_i f^j) \nu_g(E_1, \dots, E_j, \dots, E_n).$$

Since ν_g is a differential form and thus zero when its arguments are repeated, when $i \neq j$ the term in the sum is zero, so this is equal to

$$= \sum_i (E_i f^i) \nu_g(E_1, \dots, E_n) = \sum_i E_i f^i = \operatorname{div}_g X,$$

as desired.